

Simmon Yang

Richmond Hill, ON

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SUMMARY

Electrical Engineering Technology graduate (with Honours) with hands-on **electrical assembly, wiring**, and **testing** experience spanning field and bench-level work: wire/cable cutting, stripping, crimping, and termination; component-level PCB assembly and repair; and installation work performed directly from **blueprints and electrical schematics**. Precise with small parts and hand tools, comfortable following BOMs and QA checklists, and safety-trained (WHMIS, Working at Heights). Full G driver's licence, own vehicle.

TECHNICAL SKILLS

- **Assembly & Wiring:** Wire cutting, stripping, crimping & termination, custom cable/harness assembly, component-level **PCB assembly** & soldering, conduit bending & routing, mounting & securing electrical components
- **Testing & QA:** Multimeters, oscilloscopes, continuity/voltage/insulation-resistance testing, load & QA testing to verify performance and safety compliance, defect/deviation documentation
- **Drawings & Tools:** Electrical schematic & blueprint reading, BOM interpretation, AutoCAD Electrical, KiCad PCB design, hand & power tools
- **Standards & Safety:** WHMIS, Working at Heights, Electrical Code Comprehension, Lockout/Tagout

EDUCATION

Electrical Engineering Technology Diploma (with Honours)

Sep 2022 – Apr 2026

Centennial College

Scarborough, ON

- Relevant coursework: Electrical Theory, Drawing Interpretation & Installation, PLC Programming, Power Generation, Microcontrollers, Monitoring Systems, Signals & Communication
- GPA: 4.1

WORK EXPERIENCE

Electrician Assistant (As-Needed)

Sep 2024 – Apr 2026

Independent / On-call

Greater Toronto Area, ON

- Called in periodically to assist a licensed electrician on residential and commercial jobs during high-demand periods.
- Pulled, cut, routed, and terminated wiring/conduit for lighting, receptacles, and power distribution during rough-in and finishing phases.
- Installed switches, outlets, fixtures, and breakers to electrical code requirements, with correct grounding and secure device connections.
- Supported panel upgrades and device commissioning by verifying continuity, voltage, and insulation resistance before energization.

Low Voltage Technician

Jul 2025 – Sep 2025

OneNid Inc

Greater Toronto Area, ON

- Installed and terminated Cat5e/Cat6/Cat6A cabling for IP cameras, access points, and intercom systems to TIA/EIA standards.
- Mounted, aligned, and calibrated IP/analog camera systems to improve site coverage and surveillance reliability.
- Integrated access control hardware (card readers, maglocks, motion sensors) with low-voltage wiring and control panels.
- Executed structured cabling from floor plans and wiring diagrams; labeled and organized runs for easier maintenance and troubleshooting.
- Validated terminations and diagnosed faults using cable testers, crimping tools, and multimeters before system handoff.

Electrical Technician Assistant

Jul 2021 – Aug 2021

Microgreen Solar Corp

Markham, ON

- Repaired and tested DC-AC inverter power packs for solar systems under multiple load conditions.
- Assembled and terminated custom power/signal cables, validating shielding, polarity, and connector integrity.
- Performed component-level PCB troubleshooting with multimeters, oscilloscopes, and soldering tools.
- Diagnosed and resolved faults across power conversion, control, and protection circuitry.
- Executed load and QA tests to verify voltage regulation, inverter performance, and safety compliance.
- Documented repairs and test procedures to improve repeatability and maintenance traceability.

Operations & Systems Assistant

Jun 2022 – Aug 2022

Microgreen Solar Corp

Markham, ON

- Coordinated inventory and shipment logistics across product lines to support on-time fulfillment.
- Led packaging operations to consistently meet daily throughput and quality targets.

HVAC Assistant

May 2024 – Dec 2024

HVAC Contractor

Greater Toronto Area, ON

- Installed, maintained, and serviced residential/commercial HVAC systems with a focus on safe and reliable operation.
- Used wiring diagrams, schematics, and system layouts to troubleshoot installation issues and verify correct setup.

Residential Renovation Assistant

May 2023 – Aug 2023

Construction Inc

Greater Toronto Area, ON

- Installed electrical wiring for full-home renovations, including switches, outlets, and lighting fixture connections.
- Supported multidisciplinary renovation tasks (drywall, painting, demolition) to keep projects on schedule.

PROJECTS

Smart Street Lighting System

Jan 2026 – Apr 2026

Arduino Uno R3 / Embedded Systems

- Designed an Arduino Uno R3 street-lighting controller that adjusts LED brightness using ambient light, rain, and vehicle-proximity inputs, with an emergency-vehicle wave mode for priority routing.
- Produced mechanical drawings in AutoCAD and electrical schematics in KiCad; documented the design in OACETT/IEEE-style reports.

4x4x4 RGB LED Cube

Dec 2021 – Jun 2023

Arduino / C++ / PCB Design

- Built a multiplexed RGB LED cube architecture that individually addresses 192 LEDs with 6 Arduino pins using shift registers and Bit Angle Modulation (BAM), avoiding per-LED driver ICs.
- Designed custom KiCad PCBs for the LED matrix and hand-assembled/soldered all wiring and components into a 3D-printed enclosure.

Adaptive LED Headlight Simulation

Sep 2025 – Dec 2025

PIC16F747 / Embedded Systems

- Designed a microcontroller-based adaptive headlight simulation using a PIC16F747 and HC-SR04 ultrasonic sensor.
- Implemented distance-based LED control to adjust illumination zones and reduce glare exposure for oncoming vehicles.
- Tested PWM dimming behavior and identified timing constraints that introduced visible LED flicker, then corrected the control logic.

CERTIFICATIONS

- WHMIS Certification (2026)
- Working at Heights Certification (2024)
- Full G Driver's Licence (Ontario)